



HYPERLOOP IN REVIEW

January 17th, 2025

Hyperloop's evolving landscape
from January to December 2024

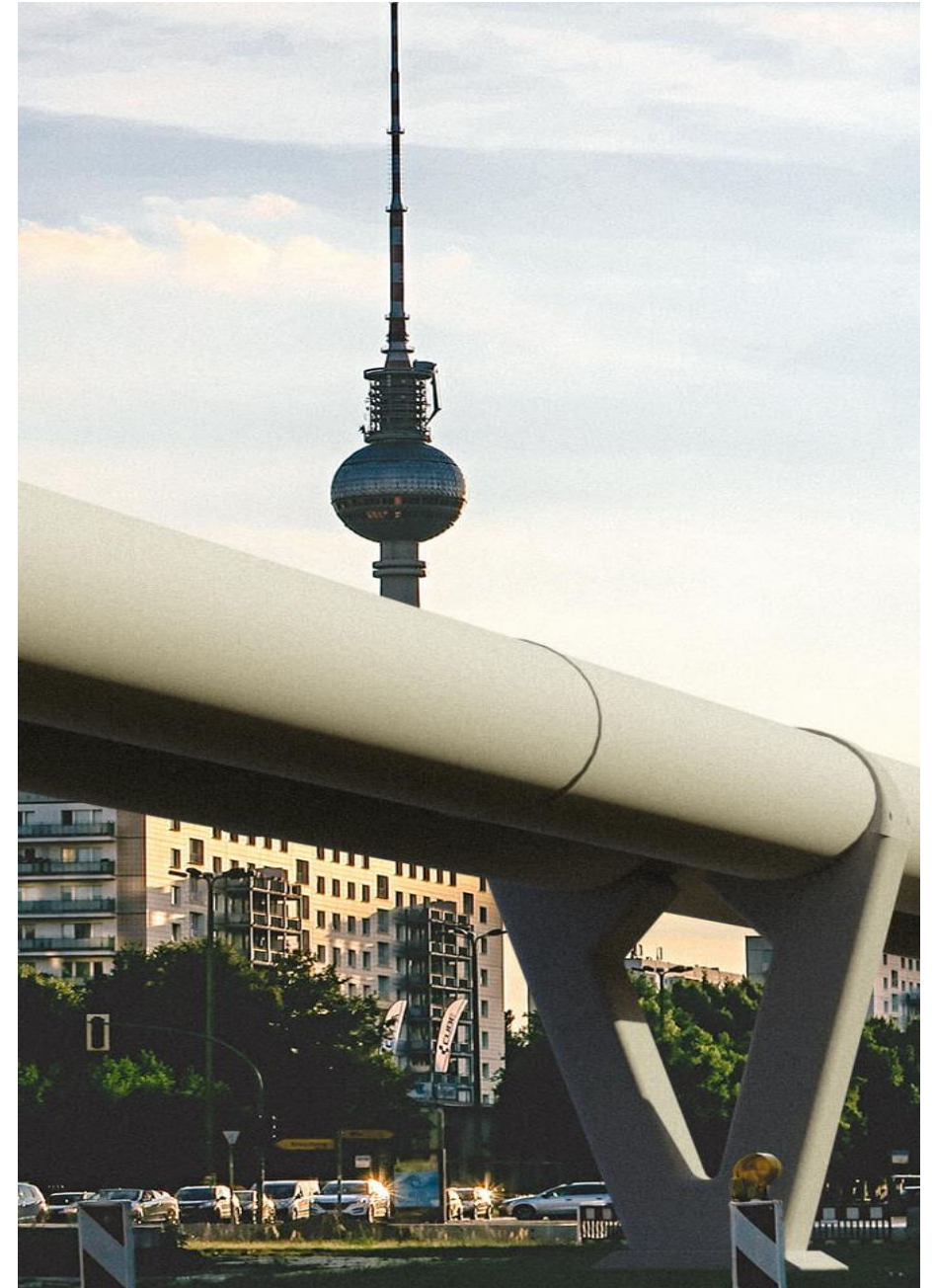
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PREFACE

Dear hyperloop enthusiast,

As 2024 has come to a close, I am filled with pride and optimism for the future of hyperloop. The past year has been nothing short of transformative - a period of growth, breakthroughs, and significant steps towards realizing a bold vision for sustainable, high-speed transportation.

Reflecting on the past year, I am excited to share the progress we have made through this annual 'Hyperloop in Review' report. Published by the Hyperloop Development Program (HDP), functioning as an observatory of hyperloop industry developments, this report encapsulates a year of remarkable advancements. It provides an overview of the milestones achieved by the hyperloop community. While it cannot capture every detail, I am eager to use this foreword to highlight the defining moments of 2024 and share my personal reflections.

2024 has been a year of dynamic change for the hyperloop ecosystem. While we faced challenges, these moments served as opportunities to recalibrate our path forward. Above all, the year has been marked by significant achievements that have solidified the foundation for the hyperloop's future:

- Strategic research infrastructures were inaugurated this year, including the DemoTube in Switzerland, goTube in Germany, and the European Hyperloop Center in the Netherlands. These facilities are now driving cutting-edge innovation and collaborative R&D efforts, placing Europe at the forefront of hyperloop technology development.

- An impressive set of 'world record' technology demonstrations show the rapid advance of hyperloop technology maturity. Amongst which: a hyperloop speed record set by CASIC in China, endurance records demonstrated by Swisspod, the first ever hyperloop through switches (entering and exiting networks) demonstrated by Hardt Hyperloop in the European Hyperloop Center as well as the full pump-down and vacuum hold performance of the European Hyperloop Center itself (which also happens to be Europe's largest vacuum chamber).
- On the political level, the hyperloop has now been officially recognized as a priority within the revised TEN-T regulation, an incredible step forward for integrating this transformative technology into Europe's future transportation network. Key European political leaders and prominent figures, including Ursula von der Leyen, Mario Draghi and Apostolos Tzitzikostas, have explicitly expressed their support for hyperloop development through policy making.
- Europe's Rail Joint Undertaking successfully awarded a proposal to establish a common hyperloop design and create a roadmap for industrialization. The Hyperloop Development Program is coordinating this project - named Hyper4Rail - with the participation of 26 other organisations from 13 countries.
- The HDP paper 'Hyperloop: Accelerating progress toward Europe's goal of sustainable transport' has been completed, bringing together the ideas of leading experts and organizations across the continent to align on a shared vision.
- The Hyper Transfer project - the first European tender for a hyperloop route - was successfully awarded in Italy to a consortium led by Webuild and Leonardo. This

milestone marks the transition from conceptual design to real-world implementation, showcasing Europe's readiness to lead in this domain.

These achievements highlight the progress made in moving hyperloop from a visionary idea to a practical reality. At the HDP, we look back on 2024 as the year the hyperloop industry took its most significant strides yet - shifting the narrative from aspirational dreams to delivering tangible results in testing and a path towards certification and industrialization.

This year has also shown us that collaboration and resilience are key. The HDP has demonstrated its ability to overcome challenges and leverage them as catalysts for a shared vision that was published in December 2024. As we look ahead to 2025 and beyond, I am confident that the hyperloop is well on its way to transforming the future of transport & mobility.

To our HDP partners, I extend my heartfelt gratitude for your invaluable contributions and steadfast support throughout this transformative year. Together, we are creating the future of transportation.

Sincerely,

*Klaus
Rudischhauser*





1. TOWARDS A NEW TRANSPORT TECHNOLOGY

Hyperloop is being developed as a high-speed guided transport and can significantly improve connections across regions and countries. Notably, being energy efficient during operations, the new transport mode will contribute to reducing energy consumption in transport and mobility on long distances.

Utilizing existing technologies

The hyperloop combines existing technologies of magnetic levitation trains and a low-pressure environment in tubes. A hyperloop vehicle can therefore move without rolling resistance and with very limited air resistance. The tubes have a low-pressure environment maintained by air pumps. This enables movement with minimal aerodynamic resistance and low energy consumption. The system can be scaled up for high-frequency transport, which is comparable to the efficiency of today's metro or light rail connections.

Enabling hybrid transportation

Hyperloop is designed for passengers and cargo. Vehicles, resembling small airplanes without wings, move through a tube. The vehicles use magnetic levitation, guidance, and propulsion, which consists of steel rails in the tube and magnets on the vehicles. Safe and efficient access for people and goods to the vehicle is being designed through applying air docks.

Enhancing multimodal solutions

Hyperloop transport requires seamless integration within a multimodal network to reach its full

potential. Transport hubs will facilitate the boarding of passengers or the transfer of cargo, as well as the transit to intercontinental aviation at airports and first- and last mile solutions within distribution centers.

Strengthening continental & regional transport

Within a European network the cruising speed is expected to be 700 km/h, according to the HDP's Hyperconnected Europe report (2022). Therefore, the hyperloop has the potential to alleviate transport over long distance, such as the strain on busy airports and congested motorways or railway networks. For different sorts of distances, the speed can be easily adjusted, making the hyperloop a versatile transportation solution for connections across a continent or in densely populated metropolitan regions.

Lowering costs with hyperloop

The hyperloop infrastructure's construction costs are comparable to high-speed rail due to its modular structure and relatively small spatial footprint, based on a CAPEX comparison executed by Deutsche Bahn E.C.O. Group and Hardt Hyperloop in 2022. In addition, the hyperloop's efficiency is increased by the friction-less magnetic propulsion and levitation as well as an enclosed environment, protecting the moving vehicles from environmental conditions. These features result in less wear and tear and, as a result, lower maintenance costs.

Energy consumption during operations

Transport mode	Per passenger kilometer (Wh)
Motorway ²	496
Aviation ¹	417
High-speed rail ²	55
Hyperloop ²	38



1. Oak Ridge National Laboratory, Transportation Energy Data Book, 2022

2. Antea Group and Hardt Hyperloop, Factsheet 'Waarom de hyperloop vanuit het stikstofvraagstuk zo interessant is', 2023





2. HYPERLOOP DEVELOPMENT IN 2024

A chronological overview of events from January until December 2024.

January

- [Webuild, Leonardo and partners](#) win the European tender for a hyperloop route – the Hyper Transfer project – in Veneto, Italy.
- [Europe's Rail Joint Undertaking](#) (EU-Rail) launches a call for proposals, which is aimed at the technical harmonization, the design concept, and roadmaps on industrialization for the hyperloop.
- [ArcelorMittal](#) pairs up with Indian partners to develop hyperloop.

February

- The [China Aerospace Science and Industry Corporation](#) successfully performs a test with hyperloop-related technology, breaking a speed record of more than 600 km/h on a 2 km test track in Shanxi, China.
- The European Hyperloop Center and the EuroTube Foundation decide to closely work together within a [European hyperloop infrastructure cluster](#) that is facilitated by the Hyperloop Development Program.
- [Klaus Rudischhauser](#), former Ambassador of the EU to Mexico, joins the HDP as Director General.

March

- The [European Hyperloop Center](#) draws global news coverage after unveiling its infrastructure that is fully equipped for testing and demonstrating.
- [Indian Railways](#) launches a tender for a demonstrative hyperloop project.

April

- The Institute of Hyperloop Technology starts the construction of the [goTube hyperloop facility](#) for which they acquired magnetic propulsion technology and power electronics from Polish hyperloop developer Nevomo.
- [Zeleros](#), the Spanish hyperloop developer, attracts new investment and embarks on a new stage by focusing on the development of batteries, electrification technologies, and transportation automation systems.
- The [Indian Institute of Technology](#) in Madras announces a hyperloop competition for 2025.



May

- In [China](#) concepts are being developed for a very first hyperloop corridor by 2035, connecting Shanghai with Hangzhou over a distance of 150 km.
- [EuroTube](#) starts the assembly of its hyperloop test facility, spanning 120 meters, in Dübendorf, Switzerland.
- [Swisspod](#) finishes the first construction phase of their hyperloop test track in Colorado, USA.

June

- The [EU](#) recognizes hyperloop as an opportunity for more sustainable technology in its transport network policy (TEN-T).
- The [German Railway Authority](#) (*Eisenbahn-Bundesamt*) initiates a research project for assessing the current state of hyperloop development. The project aims to provide objective results, including information gaps.
- Student team [Delft Hyperloop](#) demonstrates their lane switch technology at the Technological University of Delft.

Construction of DemoTube test facility in Dübendorf, Switzerland



July

- During the [European Hyperloop Week](#) in Zurich, Switzerland, university teams from around the world present prototype components and hyperloop pods to a panel of judges, competing in several categories.
- EuroTube inaugurates the first section of the [DemoTube](#), which is a hyperloop test track that is made of concrete and includes a vacuum chamber with an airlock and valve.

August

- [Hardt Hyperloop](#) conducts the first vacuum test at the European Hyperloop Center, successfully reaching the target pressure of 1 millibar.

September

- [Hardt Hyperloop](#) performs a successful test with their test vehicle, moving through the first 90 meters of the test track at a speed of 30 km/h, at the European Hyperloop Center in Veendam, the Netherlands.
- [TEKNOFEST](#), Türkiye's aerospace and technology festival, includes a hyperloop competition for magnetic levitation and suspension technologies.

October

- [Hardt Hyperloop](#) successfully tests the lane switch at the European Hyperloop Center. This technology is essential for building and operating future hyperloop transportation networks.

November

- Southwest Jiaotong University (SWJTU, Chengdu) showcases the first 300 meters of their facility which is planned to stretch over more than 1 km by summer 2025 - as shown in the picture on the right.

December

- The German [Institute of Hyperloop Technology](#) realizes the goTube testing infrastructure. The first tests are performed, which are aimed at testing propulsion systems and different vehicle concepts for freight.
- The [Hyper4Rail](#) project, supported by Europe's Rail, kicks off with a consortium of 27 partner organisations from 13 countries. The project, which is led by the HDP, aims to establish a common hyperloop design and create a roadmap for industrialization.
- Indian Railways, IIT Madras and TuTr Hyperloop achieve a milestone by completing the country's 410-meter hyperloop [test track in Chennai](#), India.





3. HYPERLOOP TEST FACILITIES IN 2024



Operational facilities

1. European Hyperloop Center (NL)
2. goTube test facility (DE)
3. ITT Madras test track (IN)
4. TUM Hyperloop test track (DE)
5. Datong Shanxi test track (CN)

Under construction

6. Chengdu test track (CN)
7. DemoTube (CH)
8. Swisspod test track (USA)

Hyperloop test track at the Indian Institute of Technology Madras in Chennai, India





4. HYPERLOOP IN POLICY MAKING

Hyperloop has been put forward in policy making, strategic plans, standardization and certification procedures, position papers and recently published studies.

Strong support from EU leadership

Ursula von der Leyen, President of the European Commission, has requested a [timetable and investment strategy](#) for the promotion and development of hyperloop technologies. Apostolos Tzitzikostas, European Commissioner for Sustainable Transport and Tourism, stressed that a [regulatory framework](#) for hyperloop must be set up. Mario Draghi highlights that the hyperloop can improve Europe's competitiveness by delivering [greater speed, efficiency, and cost savings](#).

TEN-T

The revised [Trans-European Transport Network \(TEN-T\) regulation](#) has been adopted in June 2024. The regulation now stresses that the European Commission and EU Member States should encourage projects, which aim to promote and deploy sustainable emerging technologies that enhance and ease the transport and mobility of passengers and freight. One of the examples to encourage the realization of more sustainable transportation is to include new railway technologies such as hyperloop. This recognition of hyperloop as a key technology and future transport mode highlights its potential to bring impactful added value through more sustainable,

efficient, and high-speed travel options for both passengers and freight.

Europe's Rail

In its [work plan for 2024](#), Europe's Rail identifies new approaches for guided transport modes, including magnetic levitation and vacuum tube technique, that reduce energy consumption, noise, and pollutant emissions as "an important and possibly unavoidable component of the mobility of the future."

EU regulatory framework

The Hyperloop Development Program has positioned itself on the need for an [EU regulatory framework](#) for hyperloop. Legislation for the hyperloop should be introduced from the start and can spur technological convergence of hyperloop technology on the short term, whilst ensuring cross-border interoperability and high safety standards on the long term.

Certification for passenger transport

First concepts have been approved by TÜV Süd and the Technical University of Munich (TUM) for manned operation under a framework of [hyperloop guidelines](#) aligned with the railway industry development practices and common safety methods. TÜV SÜD specifically certified the first segment of TUM Hyperloop's test tube and passenger pod.



Benelux Union

Hyperloop has been adopted in the Benelux Union [annual plan for 2024](#). The Benelux Union and its Members States have initiated a study about the potential of hyperloop development and routes, which is expected to be published in the third quarter of 2024. More specifically, the annual plan mentions possible support for a cross-border pilot project.

Saudi Arabia

The [National Strategy for Transport and Logistics](#) of Saudi Arabia has set its objectives to be reached in the railway sector, moving forward towards advancing development in the Kingdom, and contributing to increasing the GDP. The Saudi railway sector is looking at many innovations in the field of train transport technology, including high-speed trains as well as hyperloop technology.

India

The [Ministry of Railways of India](#) aspires to have more than 5 hyperloop routes by 2047 in its long-term plan. The country follows a parallel approach: 1) develop demonstration projects with global leading players, and 2) develop an indigenous system. The overall objective is to demonstrate a full-scale pilot by 2026. A global tender is to be opened.

European Hyperloop Center test facility in Veendam, the Netherlands

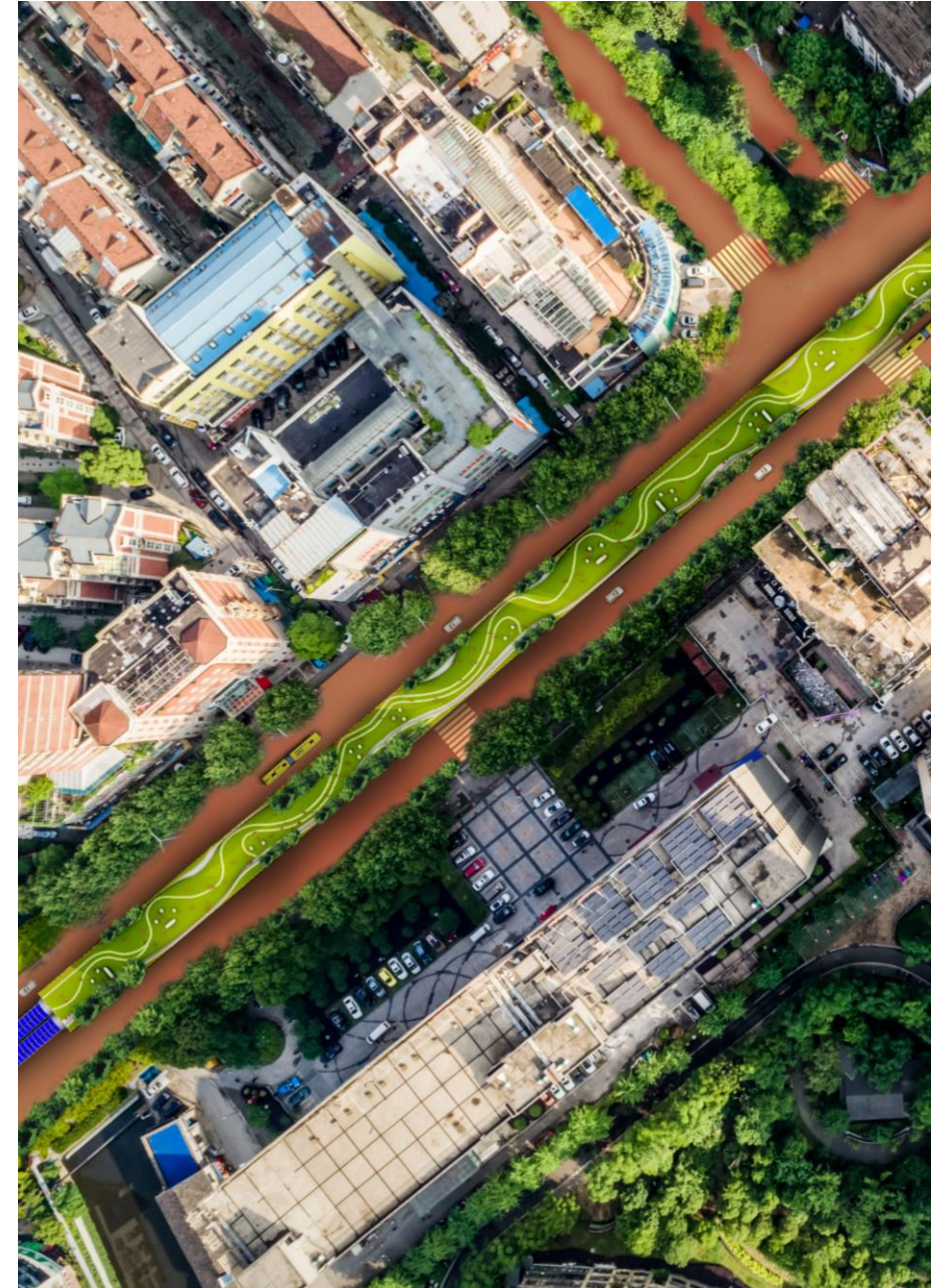




5. FUTURE OUTLOOK

In 2025, new milestones for hyperloop development are foreseen and which will lead to improved concepts being realized.

- The **Hyper4Rail** project, funded by the European Union and Europe's Rail, is expected to publish its first results in spring.
- The **European Hyperloop Week** will take place in July, during which university teams present prototype components and pods to a panel of judges, competing in several categories. This edition will be at the European Hyperloop Center in Veendam, the Netherlands.
- We hope to see a positive outcome of the **EU-led action plan for hyperloop**. We are keen on learning what the European Commission will propose for preparing next steps in the European legislative process.
- Following the initiative of the European Parliament in 2024, we expect that public funding will be made available for hyperloop research aimed at **safety of passengers**, which is an opportunity for the EU to maintain its leading role in standardization of new technologies globally.
- We look forward to the completion of the first part of the **Hyper Transfer study** in Italy, aimed at connecting the cities of Venice and Padua with hyperloop.
- We expect the **DemoTube test facility** in Dübendorf, Switzerland - of which the first section has been realized - to be expanded to a 120-meter-long tube.
- In 2025, **brand new development** projects will start, aiming to set world records and achieve groundbreaking technological advancements. These efforts are expected to drive progress in hyperloop technology, enhancing endurance, speed, stability, passenger comfort, efficiency, and cost optimization.







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